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Economic Interdependence: A Path to Peace or a Source of Interstate Conflict?*

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This article investigates the long-standing liberal hypothesis that trade ties facilitate interstate peace. Rather than assuming that trade will always promote peace, the author highlights the need to consider both the nature and context of economic linkages in assessing whether such ties are more likely to dampen or amplify interstate conflict. The study encompasses a diverse group of dyadic relationships for the period 1870–1938, 14,341 dyad years, and includes 270 militarized interstate disputes and 14 wars. After controlling for the potentially confounding influences of contiguity, regime type (joint democracy), relative capabilities, and alliance commitments, the author finds evidence that economic linkages have a dramatic influence on whether or not dyads engage in militarized disputes, but no influence on the occurrence of wars. Rather than inhibiting conflict, extensive economic interdependence increases the likelihood that dyads will engage in militarized interstate disputes. Peace through trade is most likely to arise among dyads composed of mutually dependent trading partners. Even then, the relationship between interdependence and conflict appears to be curvilinear, where low to moderate degrees of interdependence reduce the likelihood of dyadic disputes, and extensive economic linkages increase the probability of militarized disputes. Extreme interdependence, whether symmetrical or asymmetrical, has the greatest potential for increasing the likelihood of conflict.

1. *The Virtues and Vices of Trade*

Throughout history, people have debated the virtues and vices of foreign trade. For many, trade represents a path toward peace and prosperity among nations.[1] For others, trade is viewed as a contributing factor in the impoverishment of some nations and tensions between other nations. Still others view trade as largely irrelevant to leaders' decisions to engage in, or refrain from, intense forms of interstate conflict. The trade–conflict debate has long focused on whether the benefits derived through trade outweigh the costs associated with economic dependence. Those viewing trade as a mutually beneficial relationship maintain that economic ties create a disincentive to conflict. Conversely, those who focus on the negative consequences of trading relationships find the 'trade promotes peace' proposition less convincing. Despite the prevalence of trade–conflict propositions in international relations literature, there are few empirical studies investigating related propositions.

The question of the impact of trade on interstate relations is not only theoretically interesting, but has important policy implications. For centuries nations have clung to the hope that trade and interstate ties will cement the bonds of friendship that make the resort to arms unfathomable. Yet, little evidence has been provided to substantiate such claims. In light of recent trends toward regional integration, and increasing efforts to expand trade linkages among previously hostile nations, it is important to consider the potential impact of such relations.

Part of an ongoing investigation designed to enlarge our understanding about the impact of interdependence on interstate relations during the 19th and 20th centuries (Barbieri 1992, 1994a, b, 1995), this article

* Special thanks to Stuart Bremer for his assistance, valuable comments, and provision of the Correlates of War data sets used here, including the alliance, contiguity, national capabilities, and new Militarized Interstate Dispute (MID) Data set. The new MID Data were collected by the Correlates of War Project at the University of Michigan in cooperation with the Department of Political Science at Binghamton University and the Peace Science Society (International). Thanks to Andrew J. Enterline, John Camobreco, Nils Petter Gleditsch, Håvard Hegre, Jack Levy, Michael McDonald, John Oneal, Glenn Palmer, Solomon Polachek, James Lee Ray, Gregory Raymond, and Bruce Russett for their helpful comments on this research. Thanks to Brian Pollins, Philip Schafer, and J. David Singer for their assistance on the trade database project.

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begins by examining the pre-WWII period, an era that so far has received little attention. First, I consider what may be termed the unconditional liberal hypothesis – that trade promotes peace regardless of the nature and context of economic linkages. Next, I consider a proposition frequently argued by critics of commercial liberalism – that the ability of trade to promote peace is contingent upon the nature and context of economic linkages. In particular, symmetrical trading relations may foster peace, while asymmetrical dependence creates tensions that may eventually manifest themselves in conflictual interactions. Thus, both the importance (salience) and the balance (symmetry) of the trading relationship must be considered in assessing whether or not economic interdependence will inhibit interstate conflict. The study also considers alternative factors believed to alter the conflict propensity of dyadic relationships, to control for the potential confounding influence of contiguity, joint democracy, alliance commitments, and relative capabilities.

With few exceptions, economic linkages have been largely ignored in the literature devoted to the scientific study of conflict.[2] Broad historical studies of the impact of interdependence on international relations focus primarily on the system-level of analysis, with the exception of Domke's (1988) monadic level study. To date, systemic studies have failed to reach a consensus. While Domke (1988), Rosecrance (1986), and Mansfield (1994) maintain that the expansion of trade over time has resulted in a general reduction of intensive forms of interstate conflict, Waltz (1979) argues that the *decrease* in interdependence during the post-WWII period is one of a set of factors contributing to peace in that era. Apart from the lack of consensus at the system level, these studies tell us little about the dynamics operating within specific interstate relations. Just as global interdependence is believed to vary across historical periods, with differing consequences for interstate relations, it also varies in degree and character across pairs of states.

However, dyadic studies are largely confined to the post-World War II era. With the exception of Wallensteen (1973),[3] research in this area finds empirical evidence of a negative relationship between dyadic trade and conflict (Gasiorowski & Polachek, 1982; Gasiorowski, 1986a, b; Oneal et al., 1996; Polachek, 1980; Polachek & McDonald, 1992; and Sayrs, 1990)[4] Wallensteen's study spans a wider period (1920–68) and is also distinct in its identification of circumstances in which economic dependence may increase conflict (i.e., between topdogs and underdogs). In addition, Barbieri (1995) finds no evidence of an inverse relationship between trade and conflict in the post-WWII period.

Whether or not one accepts the argument that the post-WWII period is unique, it is important to determine empirically whether the trade–conflict relationship in this period is generalizable across the history of the interstate system, because the characteristics that define the post-Cold War era and relations within the evolving interstate system may be more reminiscent of earlier periods. Thus, to understand what the future may hold for interdependent relations, it is useful to begin with an examination of the past, because the notion that trade promotes peace is thought to transcend time, space, and level of analysis.

2. *Theories of Trade and Conflict*

Four propositions about the trade–conflict relationship are easily identified among a broad group of theorists: (1) the liberal argument that trade promotes peace; (2) the argument, advanced by neo-Marxists and others, that symmetrical ties may promote peace, while asymmetrical trade leads to conflict; (3) the suggestion that trade increases conflict; and (4) the belief that trade is irrelevant to conflict.

2.1 *Trade Promotes Peace*

Liberals, functionalists, and neo-functionalists argue that the expansion of interstate linkages in one area stimulates further cooperation in other areas (see Dougherty & Pfaltzgraff, 1990, ch. 10; Haas, 1958, 1964; Mitrany, 1964). Interstate linkages are thought to improve communication, reduce

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misunderstanding, and foster cultural and institutional mechanisms capable of mediating conflicts of interest that do arise. Ultimately, the recognition of mutual benefits through cooperation serves to foster peace, as national interests converge. A related economic argument attributes the inverse relationship between trade and conflict to states' recognition that trade is an increasingly more efficient means for acquiring products and markets than military conquest (Rosecrance, 1986). Finally, the argument receiving the most attention in related literature suggests that states are deterred from initiating conflict against a trading partner for fear of losing the welfare gains associated with trade (Polachek, 1980). Polachek's expected utility model of trade and conflict provides the basis for understanding both the arguments of advocates and critics of the proposition that trade promotes peace.

For Polachek and other liberals, trade patterns emerge naturally as a result of given heterogeneous factor endowments among nations. These trade patterns and the accruing benefits associated with trade affect rational leaders' foreign policy behavior, as they attempt to maximize social welfare. In a leader's expected utility calculus, the cost of conflict equals the lost welfare gains associated with potential trade losses (Polachek, 1980, 1992). Even if conflict does not lead to the cessation of trade, it will lead to inferior terms of trade, resulting in welfare losses (Polachek & McDonald, 1992). Leaders are deterred from initiating conflict against important trading partners for fear of losing welfare gains associated with trade. An extended liberal argument suggests that it is not merely the volume of trade, but the type of trade that exists between partners that affects utility calculations for conflict. Polachek & McDonald provide evidence that the more inelastic a country's import and export demand and supply to a target country, the smaller the amount of actor-to-target conflict.

While liberals recognize that gains from trade and the potential costs accompanying interdependence are not always equal, they argue that trade ties imply net positive benefits for both states. Thus, a clear link is established between expanded trade and peace within the liberal tradition. The link is tenuous for those who maintain that trade might entail net costs or for those who view states' concerns about absolute gains as subordinate to concerns about relative gains (Mastanduno, 1993; Snidal, 1993; Waltz, 1979). In addition, dependency theorists reject the notion of universal voluntary exchange, arguing that developing nations, as a result of historical-structural relationships, are not free actors and are therefore unable to make the same calculations proposed by the expected utility models (Abel & Tetrault, 1986). Thus, the existence of trade ties need not imply mutual benefits, but may reflect a lack of freedom to break free from undesirable trade relations.

2.2 *Symmetrical Trade May Promote Peace*

The notion that trade provides net benefits, which in turn reduces the likelihood of interstate conflict, has not gone unchallenged. Theoretical critiques have focused on several aspects of the unconditional liberal claim, with an eclectic group of theorists highlighting the negative consequences of economic dependence (Balogh, 1963; Cooper, 1968; Emmanuel, 1972; Gasiorowski, 1986a, b; Hirschman, [1945] 1980; Wallensteen, 1973). Although critics of liberalism seldom explicitly address the trade–conflict relationship, their commentary about the negative consequences of economic dependence is relevant to my study. For it is essentially the beneficial aspects of trading relationships that liberals assume foster peace. If costs outweigh benefits, as argued by critics of liberalism, the link of trade to peace appears less convincing. Moreover, this literature has argued that trading relations may even be a source of tension, particularly when relations are asymmetrical.

The negative consequences of dependence are believed to be greater in asymmetrical relations. The dependency school of thought and neo-Marxists have provided the most comprehensive treatment of the negative consequences of asymmetrical dependence. They argue that trade does not

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always entail net benefits, and may instead entail net costs. Rather than increasing the prosperity of trading partners, trade often results in the impoverishment of less powerful nations. A broad list of counter-claims has been advanced by dependency theorists in response to liberal claims of trade's universal benefits. These include the argument that: the gains from trade are enjoyed exclusively by developed states; trading relations between developed and developing nations retard the development process of the latter; trade destroys traditional political, economic, and social institutions; trade exacerbates inequalities in the wealth of nations; and trade relegates powerless states to a position of dependence.[5]

Moreover, dependent states may face negative political consequences when they become subject to the manipulation or coercion of the more powerful (i.e., less dependent) state in the trading relationship (Hirschman [1945] 1980). The power that arises from asymmetrical dependence may be used to gain concessions in either the political or economic domain (Hirschman; Keohane & Nye, 1977). Particularly when structural linkages exist, dependent states are less able to alter their trade patterns and may become subject to manipulation. Thus, such states will be unable to enjoy the benefits of trade without being vulnerable to coercion. When extensive economic dependence threatens national autonomy and poses problems for domestic foreign policymakers, tensions may arise among trade partners. The negative consequences of dependence are assumed to be more pronounced in asymmetrical relationships; however, all economic relationships entail costly aspects (Cooper, 1968).

Although most would agree that asymmetrical dependence serves as a source of leverage in a number of policy areas, it is unclear that the use of power arising from trading relationships is sufficient to create tensions that will manifest themselves in violent conflicts. Relations that fail to provide mutual benefits or impose disproportionate costs on one actor may be viewed as hostile relations. Yet, a hostile climate need not entail hostile acts. Hostilities might be suppressed when relations of subservience and dominance exist or when states perceive some aspect of the relationship as beneficial, and thus seek to preserve a sense of harmony. Still, it would appear plausible to argue that the existence and abuse of unequal power within asymmetrical relations creates a predisposition for conflict that is greater than that found in symmetrical trade relations. The absence of net benefits in a trading relationship neutralizes the pacifying influence of trade, assumed to exist by liberals. In fact, relations of unequal exchange may heighten tensions in interstate relations, making conflict more likely in such relationships.

Even in a climate of absolute gains for trading partners, tensions may arise over the distribution of the gains from trade. Neorealists argue that states are preoccupied with concerns over relative, rather than absolute, gains (Mastanduno, 1993; Snidal, 1993; Waltz, 1979). States may view the gain to their partner as a potential loss to themselves and will be reluctant to grant other states the benefits associated with trade. Greater wealth and productive efficiency for a trading partner may eventually be translated into military power, meaning trade creates a negative security externality (Gowa, 1994). For this reason, states may be unwilling to subordinate national interests to supranational objectives associated with trade.

2.3 *Trade Increases Conflict*

A third group of theorists, arising primarily from Marxist–Leninist, resource scarcity (Choucri & North, 1975, 1989), and neorealist thought (Waltz, 1979), envisions conflict accompanying expanded global trade. Critics also reject the notion advanced by liberals that trade will always be a desirable substitute to military options for acquiring resources and markets (Rosecrance, 1986). The history of colonialism and imperialism illustrates how military force may be used in conjunction with trading strategies to establish and maintain inequitable economic relationships (Cohen, 1973). Thus, the expansion of trade may not promote peace, but may involve increased interstate con-

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flict, as powerful states vie with one another for control over finite resources and markets (Lenin [1939] 1990), and use force to subjugate developing states to a position of dependence. Although interesting, these studies are focused at the system level of analysis and are less relevant to the study.

2.4 *Trade is Irrelevant to Conflict*

Finally, realist theorists have traditionally relegated economic relations to low politics, arguing that the influence of trade is subordinate to other considerations in determining the incidence of international conflict (Blainey, 1973; Blanchard & Ripsman, 1994; Buzan, 1984; Levy, 1989). Instead, traditional security concerns and military factors dominate leaders' calculations of the utility of conflict (Bueno de Mesquita, 1981). In addition, realists recognize that economic instruments may be an important tool in the pursuit of the national interest (Morgenthau, 1964). In such instances, trade does not represent a path toward interstate peace, but a means for securing power. Trade relations represent transitory arrangements for pursuing the national interest; when demands change, trade ties can easily be broken. In no way does trade preclude the use of alternative strategies to further the national interest, including the use of force.

One way to reconcile the differences in the propositions advanced between different theoretical traditions is to consider the linkages among the arguments presented within each school of thought. The basis of trade's pacifying effect is presumed to arise from the benefits derived from economic linkages. When such ties are believed to contribute to poverty, domestic disequilibria, or discontent over the distribution of gains from trade, the pacifying influence of trade may be neutralized. In fact, the dynamic may be reversed, where increased trade leads to increased conflict. I now turn to a way in which we might envisage the separate relationships described among the theoretical approaches in an effort to develop a set of plausible hypotheses concerning the trade–conflict relationship.

2.5 *Visualizing Interdependence*

From theories critical of liberalism, it becomes clear that not all trading relations are equal. Some trading relations may contain the necessary conditions to foster peace, while others instill hostilities or exacerbate pre-existing tensions. To understand how such variations may obtain, it is necessary to consider the nature and context of economic linkages between states. Several theorists offer clues about the types of trading relations that are most likely to produce mutual benefits and therefore possess the greatest potential for fostering peace. Theorists also provide clues about the conditions most likely to breed contempt in economic relationships, suggesting which states are less likely to enjoy the pacifying influence of trade.

One of the most important distinctions arises in the different types of relationships entailed by interdependence and dependence. Interdependence generally implies relations of mutual need and, by extension, mutual vulnerability between actors, while dependence denotes asymmetrical relations. A further distinction is made, by Keohane & Nye (1977, pp. 8–9), between interdependence and interconnectedness. Interconnectedness merely represents weak linkages among states incapable of having a significant influence on interstate relations; while interdependence entails a vulnerability and sensitivity not found in less extensive linkages.[6]

Figure 1 is helpful in identifying the conditions believed to provide the most benefits and costs of trade, simplifying the relationship between interdependent, dependent, and independent relations. Rather than thinking about the presence or absence of interdependence and dependence, as a strict dichotomy, I conceptualize trade relations along a continuum, where characteristics of different types of relationships may coexist. Figure 1 represents a dyadic relationship, where each axis measures the trade share each state has of its partner's total trade.

Quadrants I and IV represent scenarios where one state is disproportionately dependent upon the other partner. Relations of dependence are generally considered by the

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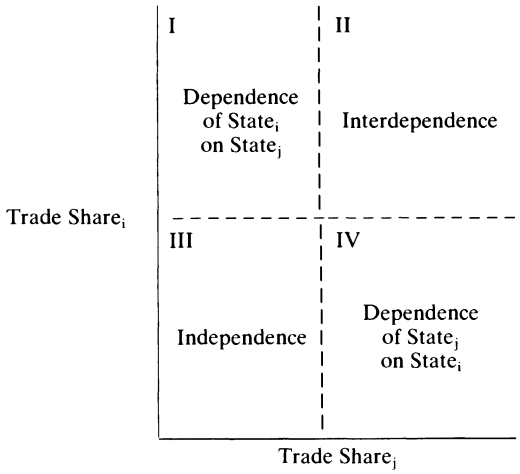
Figure 1. Dependence and Interdependence Continuum



literature to be the most conflictual. Quadrants II and III represent conditions of mutual (symmetrical) dependence, yet in the latter case dependence is minimal. Thus, dyads falling within Quadrant III have symmetrical, but not salient (i.e., important) economic relations. To distinguish between the balance and the extent of dependence, imagine a line dissecting the origin at a 45 degree angle and extending out to the point where each state has all of its trade with one partner. The closer dyads fall to the diagonal line, the more balanced the relationship. For some theorists, symmetry is the most important factor for fostering peace in a relationship regardless of the extent of such ties.[7] For others, relations falling within lower ranges of the line (i.e., in Quadrant III) lack the bonds sufficient to inhibit conflict. Generally, trading relationships that have both extensive and balanced dependence are believed to offer the greatest likelihood for peace (i.e., those falling within Quadrant II).

I now turn to an investigation of the competing propositions about the effects of economic interdependence on interstate relations. After discussing the nuances of different conceptions of interdependence, I introduce a set of measures that capture different dimensions of economic interdependence. Finally, I turn to an initial empirical investigation of the relationship between distinctive dimensions of interdependence and interstate conflict.

3. *Hypotheses*

Before turning to the empirical tests of the trade–conflict relationship, let me consolidate the alternative propositions articulated thus far in the form of hypotheses. Liberals suggest that trade promotes peace, regardless of the nature and context of interstate linkages. The greater the trade ties, the greater the deterrent to conflict. Support for the liberal hypotheses would therefore require empirical evidence that dyads composed of states with extensive economic linkages are less likely than others to engage in interstate conflict. Thus, we have the first hypothesis:

H1. Dyads composed of states with salient economic relationships are less likely than other dyads to engage in militarized conflicts.

Conversely, support for propositions advanced by critics of liberal thought would require evidence that relations characterized by symmetrical trade relations are more peaceful than asymmetrical ties. That is, symmetrical relations should promote interstate peace, while asymmetrical ties promote interstate conflict. From this, a second hypothesis emerges:

H2. Dyads composed of states with symmetrical dependence are less likely than others to engage in militarized conflicts.

From the conditions outlined in both liberal and radical schools of thought, I expect states with both mutual and extensive economic dependence to exhibit the least likelihood of interstate conflict. Such relations are assumed to provide mutual benefits, while remaining free from the conflictual elements believed to be endemic in relations of unequal exchange. From this I formulate the third hypothesis to be tested:

H3. The presence of both extensive and mutual dependence – interdependence – reduces the likelihood that dyads will engage in militarized conflict.

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Finally, while the individual dimensions of interdependence may have a significant influence on the conflict propensity of dyadic relations, salience and symmetry may have both an additive and interactive effect on conflict. The final hypothesis tests the individual and combined effects of interdependence on the occurrence of interstate conflict.

H4. The interactive and additive effects of the two dimensions of interdependence – salience and symmetry – reduce the likelihood that dyads engage in interstate conflict.

4. *Research Design*

Peace by no means requires a natural harmony of interests between states, but should at a minimum entail a commitment to seek non-military means of resolving conflicts of interest that do arise. In relation to the topic at hand, I expect interdependence to condition relationships in a manner that constrains the types of conflict behavior exhibited in interstate relations. Presumably the benefits of interdependence should make conflicts of interest less likely to arise in such relations. However, when conflicts do arise, the presence of interdependence should make military actions non-options when states choose strategies for conflict resolution.

Previous empirical studies of trade and conflict have investigated the impact of increased economic ties on a myriad of interstate relations. In this study, I focus exclusively on conflictual events, assuming that the absence of intense conflict, rather than the presence of cooperation, is more consistent with the notion of peace. Sayrs (1990) highlights the shortcomings of viewing interstate relations on a continuum ranging from cooperative to conflictual; because in the closest relationships cooperation and conflict may coexist (Coser, 1956; de Vries, 1990). Thus, to investigate whether economic linkages create a sufficient bond to deter states from resorting to threats or uses of force, I confine my analysis to the conflict propensity of dyadic relations.

4.1 *Sample and Unit of Analysis*

The unit of analysis is the dyad year, where values are recorded for the dependent variable in a given year and for the explanatory variables in the preceding year. The study is limited to sovereign states, as defined by the Correlates of War Project (Small & Singer, 1982), provided data for all variables are available for each state in the dyad. The final sample includes a total of 14,341 dyad years, for the period 1870–1938, constituting 20% of the period's population of 70,293 dyad years (excluding WWI).

Data availability poses limitations to the selection of dyads. As one would expect, information concerning interstate trade is more readily available for developed states, leading to an inherent bias toward the exclusion of developing states, and the overrepresentation of major powers in the earlier years of the study. Every effort was made to minimize this bias. There is also a bias in favor of more salient trading relationships; for when states have a small volume of trade with another state, transactions are usually recorded in an aggregated category of minor trading partners. Thus, it is impossible to distinguish between the absence of trade in a given relationship and minimal flows of trade.

4.2 *Operationalizing Variables*

4.2.1 *The Dependent Variables*

The newly revised Militarized Interstate Dispute data set is used to measure various forms of intense interstate conflict, including the occurrence of disputes and wars. Militarized Interstate Disputes (MIDs) are defined as 'a set of interactions between or among states involving threats to use military force, displays of military force, or actual uses of military force' (Gochman & Maoz, 1984, p. 586), used here as a dichotomous variable, where states composing a dyad are coded as one when they engage on opposite sides of a dispute, and zero otherwise.[8] Disputes are coded for the first year in which they occur; that is, multiple year disputes are coded only once for a particular dyad. I also investigate whether interdependent dyads are less likely to engage in the most extreme form of militarized conflict –

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wars, using the standard Correlates of War definition (Small & Singer, 1982, p. 55). The sample includes 270 militarized interstate disputes and 14 wars.[9]

4.2.2 *Interdependence*

The primary difficulty in operationalizing interdependence arises from the absence of a clear consensus about what the phenomenon entails and how it should be measured. Hirsch (1986, p. 117) identifies more than sixteen operationalizations of trade dependence in his review of two decades of dependency research. Similarly, Hughes (1971) demonstrates the impact that different conceptions and operationalizations of economic integration have on empirical findings. As such, inconsistencies in empirical research on the trade–conflict relationship may be attributed in part to these differences in operationalization.

Even among theorists who have similar conceptions of interdependence, differences exist about the preferred method of operationalizing the construct. Within the trade–conflict literature, convincing arguments have been made for employing a number of measures, including measures that: assess the importance of a dyadic flow relative to a national income or product (Oneal et al., 1996); consider the elasticity of supply and demand for commodities traded (Polachek & McDonald 1992); consider the strategic importance of commodities traded (Blanchard & Ripsman, 1994); use the value of dyadic trade to measure interdependence (Polachek, 1980); and, analyses that incorporate measures derived from both total trade and GDP (Gasiorowski, 1986a, b; Soroos, 1977). Elsewhere, I provide a comprehensive treatment of the debates over appropriate operationalizations of dependence, but here I simply employ the most desirable measures, given the data limitations for the period analyzed (Barbieri, 1995, ch. 3).

The data used to measure interdependence were obtained from a trade database constructed by the author for the pre-WWII period (see Barbieri, 1995, appendix B). Only those states for whom data were available for imports, exports, and total trade were included.[10] I measure two dimensions of dyadic interdependence, the salience and the symmetry of dependence. By salience, I mean the importance or size of the trading relationship. Low levels of salience, however equal, are not assumed to provide the necessary bonds to deter conflict between states. By symmetry, I mean the equality of dependence between partners. I argue that the interactions of these two factors, salience and symmetry, comprise what is often termed interdependence.

The indicators used to capture dependence are based upon, but are not identical to, Hirschman's ([1945] 1980, p. 85) concentration of trade index.[11] I begin by calculating the share of trade each state maintains with each partner in order to assess the relative importance of any given relationship compared to others. Trade share measures the proportion of dyadic trade over total trade, both import and export flows, for each state with its trading partners. For example, for dyadij, composed of the States i and j, i's trade share is calculated as follows:

$$(1)\ Trade\ Share_i = \frac{Dyadic\ Trade_{ij}}{Total\ Trade_i}$$

The trade shares are then used to calculate dyadic measures of salience, symmetry, and interdependence, which conform to a uniform scale that ranges from 0 to 1. Dyadic salience, calculated in Equation 2, measures the extent to which trade partners are dependent upon a given trading relationship, where high salience exists when the relationship is important for each partner.[12]

$$(2)\ Salience_{ij} = \sqrt{Trade\ Share_i * Trade\ Share_j}$$

Symmetry, calculated in Equation 3, is measured by one minus the absolute value of the difference in trade shares composing the dyad, with higher scores indicating greater equality of dependence.

$$(3)\ Symmetry_{ij} = 1 - |Trade\ Share_i - Trade\ Share_j|$$

Finally, a measure of interdependence is created, consisting of the interaction of two dimensions of economic linkages often as-

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sociated with the phenomenon – salience and symmetry. Here we are trying to assign a value that is high when both the extent and balance of dependence are high. Salience, symmetry, and interdependence have a range of values between zero and one, with mean values of 0.03, 0.9, and 0.03 respectively.[13]

(4) $Interdependence_{ij} = Salience_{ij} * Symmetry_{ij}$

4.3 *Introducing Control Variables*
Preliminary bivariate analysis identified statistically significant relationships between some of the dimensions of interdependence and interstate conflict (Barbieri, 1995, ch. 3). To assess whether these relationships might be confounded by factors affecting both trade and conflict, I control for factors consistently identified in the literature as altering the conflict propensity of dyadic relations. These factors, contiguity, joint democracy, relative capabilities, and alliance ties, are all theoretically interesting in their own right. Many of the arguments made about the control variables parallel, or are linked to, notions of interdependence. By fleshing out the separate effects of interstate bonds, I can begin to consider whether and why different types of intensive contacts may be more or less conflictual. However, in the interests of space, I focus my attention primarily on the economic linkages. For a more comprehensive evaluation of the relative impact of the control variables, see Bremer (1992b).

4.4.1 *Contiguity and Geographic Proximity*
Empirical tests have consistently revealed that contiguous dyads have both higher levels of trade (Aitken, 1973, p. 882; Arad & Hirsch, 1981) and higher levels of conflict (Bremer, 1992b; Goertz & Diehl, 1992; Gochman, 1991) than other types of dyads. Gochman (1991, p. 96) suggests that 'the interaction opportunity and potential threat to extended interests is present not only with direct, but also indirect contiguity or proximity by way of dependencies'. Thus, it is important to control for both direct and indirect contiguity.

The COW contiguity data set, as revised in 1993 by Philip Schafer, was used. Preliminary analysis (Barbieri, 1994a) indicated that there were no significant differences between different categories of contiguity; therefore, I chose to employ a dichotomous measure of contiguity, including indirect contiguity (by land and by sea up to 150 miles).

4.4.2 *Joint Democracy*
Whether or not one accepts the notion that democracies are inherently more peaceful than other states, the empirical evidence that democracies rarely go to war with each other is overwhelming (see Bremer, 1992a, b, 1993; Dixon, 1994; Maoz & Abdolali, 1989; Maoz & Russett, 1993; Morgan & Campbell, 1991; Morgan & Schwebach, 1992; Raymond, 1994; Russett, 1993). Although the causes of war and other types of militarized conflict are theoretically distinct, Bremer (1992b, 1993) provides evidence of the pacifying effect of joint democracy in less intense forms of militarized conflict.

Controlling for the influence of joint democracy is particularly important, since a high correlation is often assumed between economic (free market) and political freedom (democracy).[14] Dixon's (1994) illustration of the difficulty in disentangling the various normative and structural factors believed responsible for the joint democracy findings is compounded when we introduce trade into the analysis. Several linkages may be identified between regime type, political orientation, cultural linkages, and trade patterns, creating difficulties in separating the influence of these highly correlated variables.[15] Still, while the concepts of commercial and republican liberalism are linked, the notion that trade promotes peace is by no means contingent upon certain institutional, cultural, or political orientations. Trade is believed to offer a means through which states can transcend domestic political differences and achieve peace.

Data from Polity II (Gurr et al., 1989) are used to measure the level of democracy of each state in a dyad.[16] States with scores of five or greater on the democracy index are coded as democratic and below five as non-

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democratic. A dichotomous measure of joint democracy is employed, where dyads with two democracies are coded as one and those with one or no democracies as zero.

4.4.3 *Alliances*

Extensive economic linkages are regarded as a form of implicit (economic) alliance; yet, I must control for the presence of formal (security) alliances, which are also believed to affect the conflict propensity of dyadic relationships. This is particularly important since Gowa (1994) has shown that there is a high correlation between alliances and trade partners.

A version of the COW formal alliance data set (Small & Singer, 1969), revised by Alan Sabrosky, is used to measure alliances. Three types of formal alliances are reported in the data set: (1) mutual defense pacts, (2) neutrality agreements, and (3) ententes. I employ a dichotomous measure of alliance commitment, where the presence of any of the three types of formal alliances is coded as one and the absence of an alliance is coded as zero.[17]

4.4.4 *Relative Power*

The impact of relative power on interstate relations has received considerable attention in conflict studies, with theorists differing over whether power preponderance (Organski & Kugler, 1980) or power parity (Morgenthau, 1964) is most conducive to promoting peace. Empirical studies at the dyadic level support the power preponderance proposition (Bremer, 1993; Geller, 1993). The balance of power proposition is related, but not analogous to debates about the symmetrical nature of dependence. Although asymmetrical dependence is thought to confer disproportionate power on the less dependent state, this is not the only source of power. In addition, while relative power affects the conflict propensity of dyadic relationships, it may also affect the nature of economic dependence.

Relative power is operationalized as the relative capabilities available to each state in a given dyad. The COW Composite Index of National Capabilities (CINC) scores are used to measure each state's share of the interstate system's total military, industrial, and demographic resources (Bremer, 1980). A ratio of the larger to smaller state's capabilities is used to measure relative power.[18] The log of the relative capabilities is used in the logit regression analyses below to control for the large variations in CINC scores among states within the system. In addition, the log value has theoretical meaning, since it implies a decreasing marginal utility of increasing power differences. It makes little difference whether a state has 100 or 1,000 times more power than its opponent, since the preponderance of power should produce similar effects when power differences are great.

5. *Analyses*

In order to examine whether the presence and character of economic linkages significantly affects the conflict propensity of dyadic relations four separate logistic regression analyses are estimated for the analysis of the two dependent variables, MIDs and wars.[19] The four alternative logit models assess the impact of: (1) salience, (2) symmetry, (3) interdependence, and (4) both the additive and interactive effects of salience and symmetry – in a multidimensional model of interdependence.

5.1 *The Null Model*

Due to the rarity of the events under study – disputes and wars – any influence associated with their occurrence appears minimal. We must therefore consider not only the absolute, but the relative, effect of each variable on the probability that a dyad engage in militarized conflict. To underscore this point, I begin my analysis by estimating the null model for the samples and populations of the historical period analyzed. The null model includes only the intercept term (i.e. all β's for the explanatory variables are zero), and this allows me to calculate the mean probability that any dyad engages in a MID or war in a given year.

The mean probability that any dyad in the sample engages in a MID in a given year is approximately 0.019. That is, statistically significantly higher than the population

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Table I. Effect of Interdependence on Militarized Disputes, 1870–1938

Independent variables	Dispute occurrence$_t$			
	Model 1 Salience	Model 2 Symmetry	Model 3 Interdependence	Model 4 Full Model
Salience$_{t-1}$	3.11**	–	–	−22.64**
	(0.15)			(6.69)
Symmetry$_{t-1}$	–	−2.02**	–	−4.46**
		(0.43)		(0.80)
Interdependence$_{t-1}$	–	–	3.50**	26.60**
			(1.25)	(7.28)
Contiguity$_{t-1}$	1.67**	1.66**	1.68**	1.58**
	(0.14)	(0.14)	(0.14)	(0.14)
Joint democracy$_{t-1}$	−0.94**	−0.89**	−0.93**	−0.89**
	(0.19)	(0.19)	(0.19)	(0.19)
Alliance$_{t-1}$	−0.90	−0.09	−0.09	−0.09
	(0.21)	(0.21)	(0.21)	(0.21)
Relative capabilities$_{t-1}$	−0.19**	−0.29**	−0.18**	−0.32**
	(0.05)	(0.05)	(0.05)	(0.06)
Constant	−4.33**	−2.19**	−4.35**	0.15
	(0.15)	(0.43)	(0.15)	(0.82)
Likelihood ratio statistic (df)	273.1**(5)	284.4**(5)	272.1**(5)	301.4**(7)
Unrestricted log-likelihood[a]	−1203.5	−1197.8	−1204.0	−1189.3

Standard errors appear in parentheses.
[a] Restricted log-likelihood = −1340.0.
The restricted log-likelihood measure corresponds to the null model, while the unrestricted log-likelihood measures refer to models containing the explanatory variables.
* $p \leq 0.05$; ** $p \leq 0.01$; $N = 14341$.

mean probability of 0.007 per year for the pre-WWII period. The probability of war for the sample is 0.0009, also statistically significantly higher than the population mean of 0.0005. Thus, the conflict propensity of dyads in the sample is significantly higher than those found in the population.[20] This should be expected, since the sample is limited to dyads which have some minimal level of interaction, while the population contains both non-interacting and interacting dyads.[21]

6. *Results*

6.11 *Interdependence and Militarized Interstate Disputes, 1870–1938*

Table I exhibits the results of the first set of analyses of the impact of various dimensions of interdependence on militarized interstate disputes. In Model 1, the Salience model, I explore the question of whether important trading relationships are less conflict-prone than insignificant trading relations. Evidence in support of the liberal hypotheses would require salience to have a negative influence on the probability that dyads engage in militarized disputes. The results from Model 1 reveal that, rather than inhibiting conflict, salient trading relationships are more conflict-prone than other types of relationships, contradicting the liberal argument. Thus, the expansion of trade linkages alone seems insufficient to promote peace and may promote increased tensions in interstate relations.

Using the estimated logit model, I calculate the probability of dispute occurrence associated with various levels of economic salience, holding the control variables constant at their mean values. The probability of a dispute occurring at the minimum level of salience is 0.023, compared to 0.029 at the maximum level of salience. Thus, economically important trading partners are 1.4 times as likely to engage in a dispute than are those dyads with limited economic importance. While these findings must be interpreted with caution, since most dyads

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are clustered around the lower ranges of salience, this exercise permits me to consider the directional influence of expanded economic ties on dyadic conflict.

In Model 2, the Symmetry Model, I address the question of whether symmetrical ties inhibit conflict. Supporting evidence is revealed in Table 1, as indicated by the statistically significant and negative coefficient for symmetry ($\beta = -2.017$). This supports the notion that the balance of dependence is more important for fostering peace than the extent of trade ties. States with a similar degree of dependence are less likely to engage in disputes. Using the logit estimates, I find that the probability of a dispute when symmetry is at its minimum is 0.06, compared to 0.009 when states are equally dependent.[22] Dyads are more than seven times as likely to engage in a MID at the lowest level of symmetry than at the highest level. Thus, symmetry in exchange relations is seen to reduce the probability of militarized disputes, whereas symmetry in other forms of interstate relations (e.g., relative capabilities) is found to increase the probability of conflict.

To consider whether it is the combined effect of extensive and mutual dependence that leads to reductions in the conflict propensity of dyadic relationships, I turn to Model 3, the Interdependence Model. Surprisingly, the results from this model reveal that, rather than inhibiting conflict, high levels of interdependence increase the likelihood that dyads engage in militarized disputes. The statistically significant positive coefficient ($\beta = 3.5$) allows me to conclude with confidence that there is little evidence that those relations believed to be the most peaceful are foremost empirically. Not only does interdependence lack the strength to limit the probability of conflict, but from the findings I conclude that interdependence actually increases the likelihood of conflict. When interdependence is at its maximum, dyads are almost 25 times as likely to engage in a MID than when interdependence is at its minimum. Although these results should be interpreted cautiously, since few dyads have high values of interdependence, I can be fairly certain about the upward sloping relationship between interdependence and conflict.

Finally, the Full Interdependence Model examines the effects of the separate dimensions of interdependence and their interactive effect. Here I assume that each dimension – salience and symmetry – may independently affect conflict, but that there is also an interaction effect. By this, I mean that the growth of each condition, or the absence of either condition, may result in different effects on conflict than that which would exist if only one dimension were present. Model 4 provides evidence of the importance of considering both the additive and interactive effects of separate dimensions of interdependence. The coefficient for salience ($\beta = -22.644$) reverses its sign, from that observed in Model 1, while that for symmetry ($\beta - -4.457$) remains negative. The positive coefficient for interdependence ($\beta = 26.602$) reveals that the overall effect of the model produces a higher probability of conflict when interdependence rises. Although the salience and symmetry of the relationship initially inhibit conflict, as either dimension increases, the interaction effect reveals that the potential for conflict also increases. Thus, whether the relationship is symmetrical or asymmetrical, the extension of trade linkages will inevitably lead to an increased probability that dyads engage in militarized disputes.

To understand how the various dimensions of interdependence influence dyadic conflict, Figure 2 illustrates the overall impact of interdependence on dyadic conflict. The horizontal axis represents the trade share each state has with its partner, while the vertical axis represents the probability of a dispute occurring. The surface plane reflects the probabilities corresponding to various configurations of trade partner concentration. Overall, extensive economic linkages increase the probability of a dispute. Low levels of interdependence appear to have an insignificant effect on conflict; yet, as dyads become more dependent, the likelihood of conflict increases.

States within extreme ranges of either highly unequal or highly equal relations have the highest probability of engaging in a

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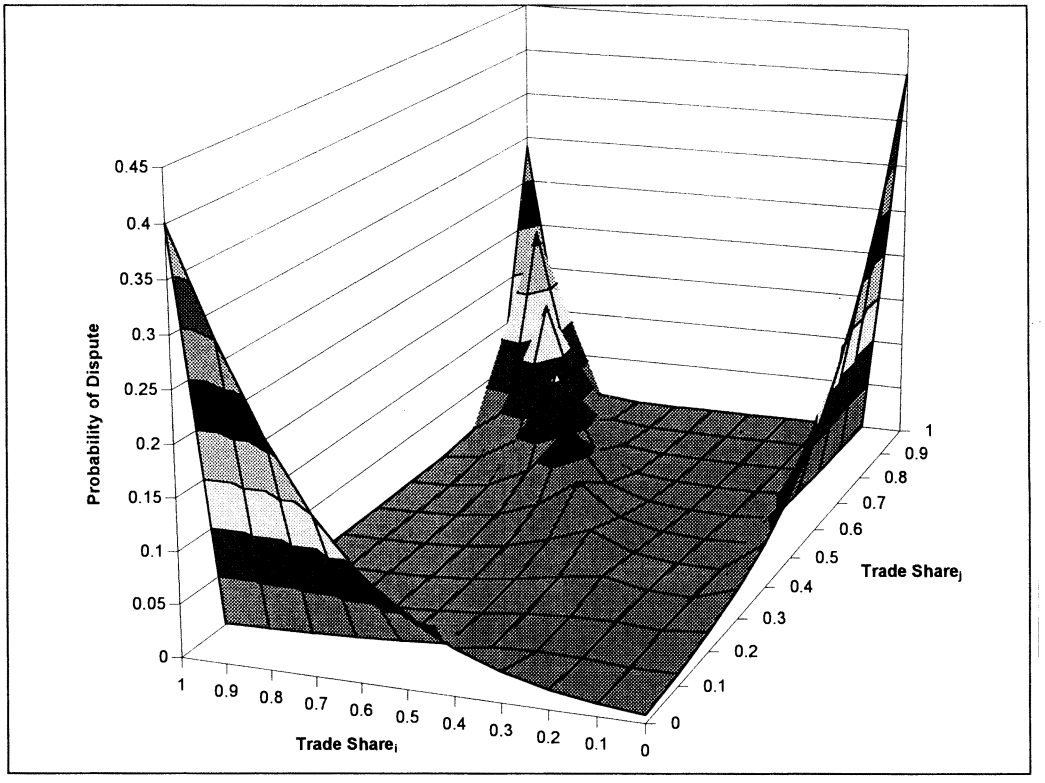
militarized dispute. This is encouraging in the sense that most of the dyads fall within the low to moderate levels of interdependence. Still, the findings suggest a cautionary note to those who view peace through integration as an indisputable outcome for the future. We might instead consider whether a hurdle effect is present in interdependent relations, whereby the benefits of such relations promote peace up to a certain point, after which the intensity of interstate relations, be they through economic or other bonds, might intensify conflict.

At most ranges, interdependence has little effect on conflict. The most adverse effect occurs among asymmetrical partners, where the probability of a dispute is 0.4. Yet, even among those with symmetrical ties, when interdependence is extensive, the probability of a dispute exceeds 0.3. Again, caution is required in interpreting the results, since the values in the middle peak are rarely observed. The relationship is extrapolated from values corresponding to dyads falling within lower ranges of symmetrical but moderately extensive ties. Still, we can imagine that the directional influence is similar, and might exist if the trend toward greater interdependence occurs in the empirical world.

6.1.1 *Control Variables*

The findings from the control variables uncover few surprises for those familiar with empirical studies of dyadic conflict. Contiguous dyads are more likely to engage in militarized disputes and democratic states are less likely to engage in militarized conflict with other democratic states. Dyads composed of states with highly unequal power are less likely to engage in conflict. Although asymmetrical dependence increases the likelihood of conflict, power derived from other forms of inequalities

Figure 2. The Impact of Multidimensional Interdependence on MID Occurrence, 1870–1938.



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appears to deter conflict. Finally, alliance commitment has no statistically significant effect on the probability that dyads engage in conflict. The results of the control variables should be interpreted with caution, since the sample does not reflect a random selection process, but they are consistent with previous findings for larger samples (e.g., Bremer, 1992b).

We can consider the goodness of fit for each alternative model in comparison to the null model (i.e., where the β's for all explanatory variables equal zero) by using the likelihood ratio test.[23] The likelihood ratio test assesses whether the inclusion of the explanatory variables provides a significant improvement over the null model for explaining MID occurrence. In Table I we see that the likelihood ratio statistic is statistically significant for each model, revealing the importance of considering economic interdependence and the control variables when seeking to uncover missing links in the conflict puzzle.

6.2 *Interdependence and War*

How does interdependence affect war, the most intense form of conflict? Table II gives the empirical results. The rarity of wars makes any analysis of their causes quite difficult, for variations in interdependence will seldom result in the occurrence of war. As in the case of MIDs, the log-likelihood ratio tests for each model suggest that the inclusion of the various measures of interdependence and the control variables improves our understanding of the factors affecting the occurrence of war over that obtained from the null model. However, the individual interdependence variables, alone, are not statistically significant. This is not the case with contiguity and relative capabilities, which are both statistically significant. Again, we see that contiguous dyads are more conflict-prone and that dyads composed of states with unequal power are more pacific than those with highly equal power. Surprisingly, no evidence is provided to support the commonly held proposition that democratic states are less likely to engage in wars with other democratic states.

The evidence from the pre-WWII period provides support for those arguing that economic factors have little, if any, influence on affecting leaders' decisions to engage in war, but many of the control variables are also statistically insignificant. These results should be interpreted with caution, since the sample does not contain a sufficient number of wars to allow us to capture great variations across different types of relationships. Many observations of war are excluded from the sample by virtue of not having the corresponding explanatory measures. A variable would have to have an extremely strong influence on conflict – as does contiguity – to find significant results.

7. *Conclusions*

This study provides little empirical support for the liberal proposition that trade provides a path to interstate peace. Even after controlling for the influence of contiguity, joint democracy, alliance ties, and relative capabilities, the evidence suggests that in most instances trade fails to deter conflict. Instead, extensive economic interdependence increases the likelihood that dyads engage in militarized dispute; however, it appears to have little influence on the incidence of war. The greatest hope for peace appears to arise from symmetrical trading relationships. However, the dampening effect of symmetry is offset by the expansion of interstate linkages. That is, extensive economic linkages, be they symmetrical or asymmetrical, appear to pose the greatest hindrance to peace through trade.

Although this article focuses exclusively on the pre-WWII period, elsewhere I provide evidence that the relationships revealed here are also observed in the post-WWII period and more extended period, 1870–1985 (Barbieri, 1995). Why do the findings differ from those presented in related studies of the trade–conflict relationship, which reveal an inverse relationship between trade and conflict? Several explanations, other than the temporal domain, can be offered. First, researchers differ in the phenomena they seek to

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Table II. Effect of Interdependence on Wars, 1870–1938

Independent variables	War occurrence$_t$			
	Model 1 Salience	Model 2 Symmetry	Model 3 Interdependence	Model 4 Full Model
Salience$_{t-1}$	−0.97	–	–	−43.87
	(5.25)			(55.20)
Symmetry$_{t-1}$	–	0.70	–	−3.52
		(2.94)		(5.19)
Interdependence$_{t-1}$	–	–	−0.19	46.80
			(6.13)	(58.13)
Contiguity$_{t-1}$	2.60**	2.60**	2.57**	2.54**
	(0.79)	(0.78)	(0.79)	(0.79)
Joint democracy$_{t-1}$	−1.33	−1.36	−1.36	−1.34
	(1.05)	(1.04)	(1.05)	(1.06)
Alliance$_{t-1}$	0.03	0.02	0.02	0.05
	(0.78)	(0.78)	(0.78)	(0.78)
Relative capabilities$_{t-1}$	−0.61*	−0.59*	−0.61*	−0.64*
	(0.28)	(0.29)	(0.28)	(0.30)
Constant	−7.23**	−7.95**	−7.25**	−3.82
	(0.81)	(3.04)	(0.81)	(5.21)
Likelihood ratio statistic (df)	30.5**(5)	30.5**(5)	30.4**(5)	31.3**(7)
Unrestricted log-likelihood[a]	−95.8	−95.8	−95.8	−95.4

Standard errors appear in parentheses.
[a] Restricted log-likelihood = −111.04.
The restricted log-likelihood measure corresponds to the null model, while the unrestricted log-likelihood measures refer to models containing the explanatory variables.
* $p \leq 0.05$; ** $p \leq 0.01$; $N = 14341$.

explain, with many studies incorporating both conflictual and cooperative interstate behavior (e.g., Gasiorowski, 1986a, b; Gasiorowski & Polachek, 1982; Polachek, 1980, 1992; Polachek & McDonald, 1992). Studies that focus exclusively on extreme forms of conflict behavior, including disputes and wars, differ in their spatial and temporal domains, their level of analysis, and their measurement of central constructs. Preliminary tests reveal that the composition of dyads in a given sample may have a more dramatic impact on the empirical findings than variations in measurement. For example, the decision to focus exclusively on 'politically relevant dyads' may be one source of difference (Oneal et al., 1996).

Perhaps the primary component missing from this and related research is the inclusion of a more adequate assessment of the costs and benefits derived from interdependence. I have repeatedly argued that the conflictual or pacific elements of interdependence are directly related to perceptions about trade's costs and benefits. Yet, a more comprehensive evaluation of these costs and benefits is needed to see whether a link truly exists between the benefits enjoyed in a given trading relationship and the inhibition of conflict in that relationship, or conversely, the presence of net costs for at least one trading partner and the presence of conflict in that relationship. For example, are trading relationships that contain two partners believed to benefit from trade less conflict-prone than those containing at least one partner perceived to be worse off from trade? I have merely outlined the types of relationships believed to confer the greatest benefits, but such benefits and costs require a more rigorous investigation.

In addition, while liberals generally assume benefits increase with the expansion of trade, benefits may actually conform to the law of diminishing returns. Costs, on the other hand, may grow exponentially, as interdependence grows. Thus, if we evaluate the net benefits of interdependence, costs may outweigh benefits as interdepen-

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dence increases. Again, further exploration is needed to evaluate the functional form that conforms to the rise and/or decline of costs and benefits in interdependent relations. Of course, assessing the political costs of interdependence may be more difficult than assessing economic benefits. For example, it is easier to evaluate the variations in national growth corresponding to greater participation in trade than to assess the political impact of interdependence, such as losses in national autonomy.

Another area that requires further consideration concerns the interactive effects of several types of interdependencies for fostering interstate peace. One of the primary components of liberal theory focuses on the convergence of interests and transmission of commonalties that arise through economic and non-economic linkages. These norms are believed to facilitate the emergence of formal and informal mechanisms to mediate conflict, when conflicts of interest do arise. This issue has been the subject of attention in the democratic peace literature, but remains unexplored in studies of economic interdependence. Yet, one could easily argue that an interactive effect might be present in the expansion of different types of interstate linkages for facilitating mechanisms conducive to conflict resolution. For example, the presumed norms of conflict mediation present in jointly democratic dyads might facilitate the resolution of conflict, when economic incentives create the impetus for such mediation. In other words, even if actors have an economic incentive to seek non-violent strategies to resolve conflicts of interest, the absence of sufficient mechanisms to offset conflict may make conflict more likely to arise. The issue of an interactive process between different types of interdependencies leads to another area of research that requires future exploration, the countervailing influences of different forms of interstate interdependencies.

What is clear from related empirical research is that commonalities and various types of interdependencies have different effects on interstate conflict. Contiguity is consistently found to increase the likelihood of conflict, while joint democracy is found to inhibit conflict. When tracing the rationale for these findings, theorists often use the concept of interstate ties to drive their explanation. Moreover, some theorists argue that extensive linkages found among contiguous states is a contributing factor to their high conflict propensity, while at the same time suggesting that linkages and commonalities found among jointly democratic dyads contributes to their pacific relationships. One might therefore conclude that extensive linkages foster both peace and conflict. How do we determine the conditions under which one of these effects is more likely to arise in a given interdependent relationship? Clearly, there must be some underlying factors that influence the tendency for some forms of interdependence to inhibit conflict, while others increase conflict. That is, it is not interdependence itself that determines the impact on interstate relations, but some characteristic of the type of interdependence present in the relationship.

While I sought to answer several questions about the trade–conflict relationship, at the very least it should be clear that interdependence is a more complex relationship than generally depicted by theorists. It is also clear that economic ties have a significant influence on conflict. Although interdependence's impact on conflict is not that envisioned by liberals, the tendency for conflict studies to exclude this important relationship can no longer be accepted. Economic interdependence affects interstate relations. While it may not promote interstate peace, it is essential to understanding the puzzle of interstate conflict. Clearly, the time has come to elevate economic relationships from their traditional secondary status to center stage in the study of international relations.

NOTES

1. For a comprehensive history of this and related economic thought, see Spiegel (1991); see de Wilde (1991) for a history of the interdependence tradition in political science.
2. For comprehensive reviews of the scientific study of conflict, see Bremer (1992b), Bremer & Cusack (1993), and Vasquez (1993).

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3. Focusing on the post-WWII period, Russett (1967) also provides evidence that increased trade leads to increased conflict, but his study is concerned with intra-regional trade and conflict.
4. Other scholars provide evidence of an inverse relationship between trade and political climate and/or orientation at the dyadic level, but view causation as flowing from politics to trade patterns (Dixon & Moon, 1993; Pollins, 1989a, b). The potential for simultaneity here is obvious, but will be analyzed at a later date. Polachek (1992) provides evidence that the effect of trade on conflict is greater than that of conflict on trade. To reduce the confounding influence that intense warfare has on trade patterns, my study excludes the WWI period.
5. For a comprehensive review of the dependency school of thought, see Blomström & Hettne (1984).
6. 'Sensitivity refers to the openness of a country to changes in other countries that are transmitted by a mutual interaction, vulnerability refers to the effects of changes in rules or policies' (quote from Gasiorowski, 1986a, p. 24; see Keohane & Nye 1977, pp. 12–13).
7. Advocates of South–South economic unions may view less significant, but equal, economic linkages as a greater force for peace than extensive relations under the guise of unequal exchange. According to this view, the linkages among developing states would eventually expand, providing increasing benefits (economic and political) in the long run and breaking the cycle of dependence on developed states.
8. Since the dependent variable, MID occurrence, is really an event count, a case could be made that an event count model should be used in this analysis. I chose to dichotomize the occurrence of at least one event, since the probability that any dyad would experience a MID more than once is extremely rare (e.g., only 11 cases were identified in the sample), and the occurrence of at least one MID is sufficient to violate the conditions of peace.
9. Wars are a subset of MIDs and are included in the MID analysis.
10. For a comprehensive discussion of the trade database and a complete listing of sources, see Barbieri (1995, Appendix B). Sources of trade data include League of Nations (1910–40); *Statesmen's Yearbook*; and Mitchell (1982, 1983). Data were collected in local currency and converted to US dollars using exchange rates from Bidwell (1970); Gurr et al. (1989); *Statesmen's Yearbook (1870–1940)*; and the US Department of Commerce.
11. Hirschman notes that 'Examining the conditions under which country A will experience most difficulties in shifting its trade from one country B to other countries, we found that the fraction which B holds in A's total trade is an important element in evaluating the situation. This fraction depends on the size of its numerator, the trade of B with A, and on the size of its denominator, A's total trade' ([1945], 1980, p. 85; see also pp. 30–31).
12. The geometric, rather than arithmetic mean, is used, since the former assigns higher values to dyads where each state has high trade shares and lower values, than those derived from the arithmetic mean, when trade shares are highly unequal.
13. Although the dimensions of interdependence appear to be related, they are considered distinct phenomenon. Correlation tests reveal that salience and symmetry are only moderately correlated (r = 0.55). However, a high correlation exists between salience and the interaction term, interdependence (r = 0.90). Despite the high correlation, I include interdependence as a separate variable, since preliminary tests reveal that it is statistically significant and it is assumed to be theoretically distinct from salience. See Friedrich (1982) for a useful discussion of the guidelines for including interactive variables. No other variables used in the study were found to be collinear. For correlation tests among the control variables, and between the control variables and the interdependence measures, see Barbieri (1995, Appendix A).
14. Rummell (1983, 1985) classifies states as libertarian only when domestic economic freedoms exist. We may assume that states with domestic economic freedoms are more likely to promote free trade policies, which leads to difficulties in separating the precepts of commercial and republican liberalism.
15. Surprisingly, correlation tests reveal no problem with multicollinearity among the control variables and the interdependence measures.
16. 'A state is classified as democratic if its chief executive is directly or indirectly elected in a popular fashion and its legislative branch is also elected and able to constrain the executive in an effective manner' (Bremer, 1992b, p. 9).
17. Bremer (1993, p. 237) suggests that a dichotomous measure of alliance ties is sufficient to capture the effect of alliances on conflict, although variations exist on the effect of different alliance types.
18. A measure of overwhelming preponderance (Weede, 1976) was also employed in preliminary tests, with no significant differences observed in the results.
19. For information about logit regression analysis and interpretation of results from tests conducted, see Aldrich & Nelson (1984) and King (1989).
20. For a discussion of the tests used to compare the sample and population means, see Hayslett (1968, p. 86).
21. Håvard Hegre (letter, 25 April 1995) suggests that the high conflict propensity found in the sample may be explained, in part, by the overrepresentation of major powers, which tend to be more conflict-prone than other states. The bias arises not only from the greater availability of data for major powers, but also from their tendency to conduct trade with more partners than do minor powers. Thus, they will be represented in more dyadic relationships.
22. The estimated logit models are used to calculate the dispute probabilities, while holding the control variables constant at their mean values.

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23. Comparing the restricted and the unrestricted models (i.e., the null model versus the model incorporating the explanatory variables) allows us to evaluate whether the inclusion of the explanatory variables improves our understanding of conflict over that provided by the null model. When the likelihood ratio is significant, we may conclude that the unrestricted model provides a better explanation of conflict than that found in the null model. The likelihood ratio test statistic is computed as follows: $\mathfrak{R} = [2(\ln L_{unrestricted}{}^{*} - L_{Restricted})]$, where $\mathfrak{R}$ is chi-squared distributed with degrees of freedom equal to the number of parameters of the null model (see King, 1989, pp. 84–87). In Tables I and II, a significant likelihood ratio statistic, $\mathfrak{R}$, means that the unrestricted model improves upon the explanatory power of the null model. LIMDEP Version 5.1 is used for this and other estimations performed here.

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APPENDIX A. Descriptive Statistics, 1870–1938

Variable	Mean	SD	Min	Max	N[a]
Year	1916.6	18.929	1871	1938	14341
Dyadic trade[b]	35.414	90.107	0.0001	2872	14341
Total trade$_i$[c]	1703.3	3412.5	0.1049	150800	14341
Total trade$_j$	1031.1	1902.6	0.4985	58960	14341
CINC$_i$[d]	5.3158	7.2063	0.006	29.05	14341
CINC$_j$	3.3776	4.5669	0.006	23.72	14341
Democracy$_i$[e]	5.6619	3.7105	0	10	14341
Democracy$_j$	4.6851	3.7952	0	10	14341
Joint democracy[f]	0.2564	0.4367	0	1	14341
Contiguity[g]	0.268	0.443	0	1	14341
Alliance[i]	0.0561	0.2301	0	1	14341
HiAct[i]	0.2257	1.8822	−9	22	14341
Mid[k]	0.0188	0.1359	0	1	14341
War[k]	0.001	0.0312	0	1	14341
Capabilities ratio[l]	46.676	164.93	1	4501	14341
Log (LS ratio)[m]	2.2996	1.5913	0.0001	8.412	14341
Trade Share$_i$[n]	0.041	0.079	0.0000	0.993	14341
Trade Share$_j$	0.07	0.122	0.0000	0.983	14341
Salience[o]	0.0331	0.0428	0.0001	0.9685	14341
Symmetry[p]	0.9198	0.1219	0.0374	1	14341
Interdependence[q]	0.0278	0.0338	0.0001	0.9569	14341

Subscripts *i* and *j* correspond to State$_i$ and State$_j$, respectively.
The variables corresponding to State$_i$, explained below, are similarly converted for State$_j$.
Pre-WWII trade data are derived from database created by the author (see Barbieri, 1995).
Conflict measures are from the new Militarized Interstate Dispute (MID) data set.
Alliance, contiguity, and CINC scores are from the COW Project.
[a] Refers to sample after introducing lag variables and excluding cases with missing values.
[b] Dyadic trade, excluding zero trade flow values.
[c] Imports plus Exports for State$_i$.
[d] Composite Index of National Capabilities (CINC) scores, COW Project.
[e] Democracy score for State$_i$, from *Polity II* (Gurr et al., 1989).
[f] Joint Democracy = 1 when Democracy$_i$ and Democracy$_j$ > = 5, and zero otherwise.
[g] Contiguity = 1 when direct or indirect contiguity is present and zero otherwise.
[h] Alliance = 1 when any type of alliance is present, zero otherwise.
[i] Highest level of force in a dispute, coded 1–22 (−9 indicates missing data).
[j] Mid = 1 when dyads engage on opposite sides of a dispute, and zero otherwise.
[k] War = 1 when dyads engage on opposite sides of a war, and zero otherwise.
[l] Larger to smaller CINC score ratio.
[m] Log of the Capabilities Ratio.
[n] TS_i = Dyadic Trade/Total Trade$_i$.
[o] Salience = SQR ($TS_i \times TS_j$).
[p] Symmetry = 1 − ABS ($TS_i - TS_j$).
[q] Interdependence = Salience$_{ij}$ × Symmetry$_{ij}$.

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